1 When a current passes through a wire, the wire becomes hot and expands.

For Examiner's Use

This can be investigated in a laboratory by passing a current through a wire of diameter d and measuring the displacement y, as shown in Fig. 1.1.

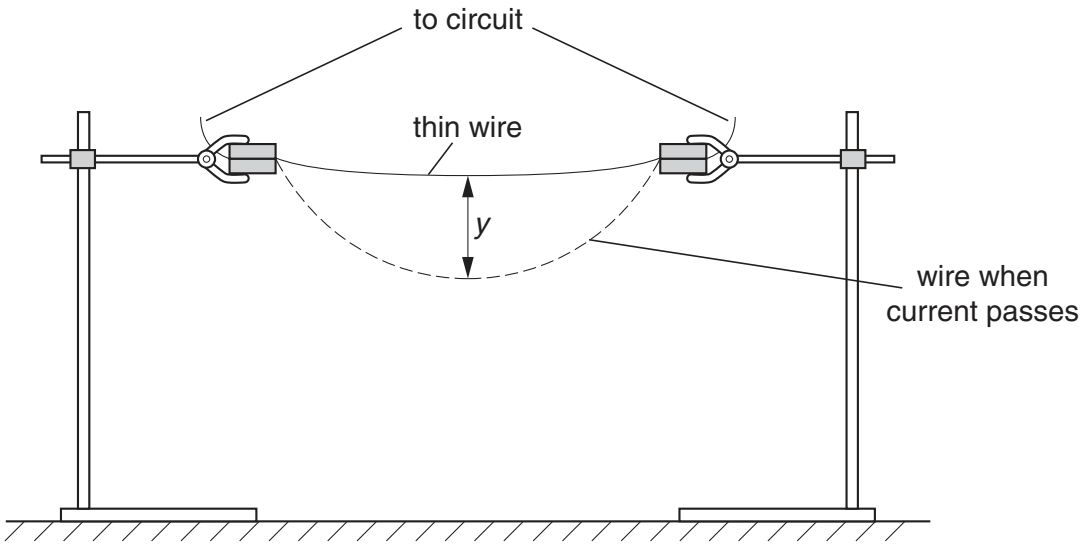

Fig. 1.1

It is suggested that the diameter d of the wire is related to y by the equation

$$y = pd^q$$

where p and q are constants.

Design a laboratory experiment to investigate the relationship between d and y, so as to determine a value for q. You should draw a diagram showing the arrangement of your equipment. In your account you should pay particular attention to

(a) the procedure to be followed,

(b) the measurements to be taken,

(c) the control of variables,

(d) the analysis of the data,

(e) the safety precautions to be taken.

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